# D2.3 Water potential (HL)

NAME

DATE

**How to use this sheet:** go through the slides and videos with it, then close them and write from memory. Use your own words and add page numbers from the book (pp. 690–692) where you want more detail.

**Guiding questions.** What affects whether water moves into or out of a cell? In what ways do plant cells and animal cells control water movement differently?

## D2.3.8–9 Water potential and the direction of movement

Definition, unit, the value for pure water, and why solutes make it negative. Which way does water move?

## D2.3.10–11 Solute and pressure potential

Write the equation. Fill in the table.

| | SOLUTE POTENTIAL | PRESSURE POTENTIAL | WATER POTENTIAL |
|---|---|---|---|
| Cell in a hypotonic solution | | | |
| Cell in a hypertonic solution | | | |

How to estimate the water potential of potato tissue:

## Before you finish

Tick the ones you can explain without notes. Circle anything to ask about.

- ☐ Water potential
- ☐ Pure water = 0
- ☐ Direction of movement
- ☐ Solute potential
- ☐ Pressure potential
- ☐ Turgid and flaccid
- ☐ Plasmolysis
- ☐ Potato practical

What the Cell. Original material written for this course.